PAPER_297 — UQFF Superluminal Hubble Expansion Ratio η_exp = 3.328 > 1

Author: Daniel T. Murphy **Date:** March 17, 2026 ## First UQFF Module Where v_exp/c > 1 at System Boundary

Session: 84
Module: `UNIVERSE_DIAMETER_UQFF_MODULE.cpp` (26th C++ UQFF module — Observable Universe as System)

Classification: Unique Physics — First UQFF Superluminal Expansion Parameter (η_exp > 1)

Abstract

The UQFF Observable Universe Diameter Module is the **first UQFF module** where the system boundary recession velocity exceeds the speed of light: `v_exp = H₀ × r_obs = 9.984×10⁸ m/s = 3.328c`. The dimensionless Superluminal Expansion Ratio η`_exp = v_exp/c = 3.328 > 1` is a new UQFF parameter encoding the cosmological property that the observable universe spans **3.328 Hubble lengths** (`r_obs = 3.328 × r_H`). The Hubble-expansion coupling factor at t_Hubble is `(1 + H₀ × t_H) = 1.988` — a near-doubling of the Newtonian base over cosmic time. All 25 prior UQFF modules had η_exp << 1 (sub-luminal expansion).

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1. Physical Setup

System: Observable Universe
Observable universe radius: r_obs = 4.4×10^{26} m
Hubble constant: H_0 = 70 km/s/Mpc = 2.269×10^{-18} s^{-1}
Speed of light: c = 3.0×10^{8} m/s
Hubble sphere (Hubble radius): r_H = c/H_0 = 1.322×10^{26} m = 4.28 Gly
Cosmic age: t_H = 4.355×10^{17} s = 13.8 Gyr

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2. Master Relation

Hubble recession velocity at observable universe boundary:

$$v_{exp} = H_0 \times r_{obs} = 2.269 \times 10^{-18} \times 4.4 \times 10^{26} = 9.984 \times 10^8 \text{ m/s}$$

Superluminal Hubble Expansion Ratio:

$$\boxed{\eta_{exp} = \frac{v_{exp}}{c} = \frac{9.984 \times 10^8}{3.0 \times 10^8} = 3.328 > 1}$$

Hubble length ratio:

$$\frac{r_{obs}}{r_H} = \frac{v_{exp}}{c} = \eta_{exp} = 3.328$$

The observable universe boundary is 3.328 Hubble lengths from Earth — comfortably beyond the Hubble sphere where recession velocity equals c.

3. Hubble-UQFF Near-Doubling of Gravity

The base gravity term with Hubble expansion coupling:

$$a_{base}(t) = g_{base} \times (1 + H(z) \times t) = 3.447 \times 10^{-10} \times (1 + 2.269 \times 10^{-18} \times t)$$

At t = t_H = 4.355×10^{17} s:

$$a_{base}(t_H) = 3.447 \times 10^{-10} \times (1 + 0.988) = 3.447 \times 10^{-10} \times 1.988 = 6.854 \times 10^{-10} \text{ m/s}^2$$

Hubble expansion factor ξ`_H = 1 + H`$_0$` × t_H = 1.988`:

This near-doubling factor (≈2) reflects that the UQFF base gravity **almost doubles** over the Hubble time when the Hubble coupling is included — a striking result confirming that the Universe-scale Hubble coupling is an O(1) effect (not a small correction).

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4. Superluminal Expansion in the EM Lorentz Term

The EM Lorentz term explicitly incorporates η_exp:

$$a_{EM} = \frac{q \cdot v_{exp} \cdot B_{cosmic}}{m_p} \times (1 + \eta_{exp}) \times \text{scale}_{EM}$$

With η`_exp = 3.328`, the factor `(1 + `η`_exp) = 4.328` vs. `(1 + 1) = 2` for a subluminal system. This introduces a **42% EM enhancement** relative to a hypothetical c-speed boundary reference.

Numerically:

$$a_{EM} = \frac{1.602 \times 10^{-19} \times 9.984 \times 10^{8} \times 10^{-15}}{1.673 \times 10^{-27}} \times 4.328 \times 10^{-12}$$

$$= 95.59 \times 4.328 \times 10^{-12} = 4.136 \times 10^{-10} \text{ m/s}^2$$

The EM term (4.136×10^{-10} m/s^2) is **comparable to the Newtonian base** (3.447×10^{-10} m/s^2) — another first for UQFF modules.

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5. Special Relativity Compatibility

The superluminal recession velocity `v_exp > c` does NOT violate special relativity. This is a **coordinate velocity** (metric expansion), not a proper velocity between local inertial frames. In GR, the expansion of the universe allows coordinate distances to grow faster than c, as confirmed by the cosmological horizon structure. The Hubble-flow velocity η_exp > 1 is part of the cosmological metric, not a violation of local Lorentz invariance.

Specifically, this corresponds to objects beyond the **Hubble sphere** (r > r_H) in comoving coordinates. For the observable universe at r = 4.4×10^{26} m with r_H = 1.322×10^{26} m, we are 3.33 Hubble lengths out — solidly in the superluminal expansion regime.

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6. η_exp Parameter in UQFF Architecture

Module	Session	r_obs (m)	v_exp/c (η_exp)	η_exp > 1
SGR1745	65	~0.01 pc	~0 (local)	No
Pillars of Creation	68	3.3×10^{17}	$\ll$1	No
NGC1792	73	7.6×10^{20}	$\ll$1	No
Andromeda	75	1.04×10^{21}	$\ll$1	No
HUDF (z=3.5)	72g	1.23×10^{27}	~9 (co-moving)	Yes (but not UQFF param)
Universe Diameter	**84**	**4.4×10^{26}**	**3.328**	**Yes — FIRST explicit**

The key distinction: prior modules used H(z) as a correction factor, never explicitly computing or naming η_exp as a parameter. PAPER_297 establishes η_exp as a first-class UQFF parameter.

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7. Hubble Sphere as UQFF Speed-of-Light Boundary

Define the **UQFF Hubble Horizon** as the radius where η`_exp = 1`:

$$r_H = \frac{c}{H_0} = \frac{3 \times 10^8}{2.269 \times 10^{-18}} = 1.322 \times 10^{26} \text{ m} = 4.28 \text{ Gly}$$

Objects beyond r_H recede superluminally. The observable universe (r_obs = 4.4×10^{26} m) extends to 3.328 r_H — confirming we can observe objects that are currently receding faster than light (their photons from early epochs can still reach us).

This adds a new entry to the UQFF speed-of-light boundary catalog: - **PAPER_264**: Anti-gravity boundary at f_TRZ = -1 (HUDF module) - **PAPER_266**: Meissner quench at B = B_crit - **PAPER_297**: Hubble superluminal horizon at r = r_H (**this paper**)

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8. WOLFRAM Term

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eta_exp=v_exp/c=H0*r/c=9.984e8/3e8=3.328>1;
FIRST UQFF eta_exp>1;
r_H=c/H0=1.322e26m Hubble sphere;
r_obs=3.328*r_H;
expansion_factor(t_H)=1+H*t=1.988(near-doubling);
EM term a_EM prop eta_exp [PAPER_297]
```

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9. Key Values Summary

Quantity	Symbol	Value	Unit
Hubble recession velocity	v_exp	**9.984×10^8**	m/s
Superluminal ratio	η_exp	**3.328 > 1**	dimensionless

Quantity	Symbol	Value	Unit
Hubble radius	r_H	1.322×10^{26}	m
Observable / Hubble ratio	r_obs/r_H	3.328	dimensionless
Hubble expansion factor	ξ_H	**1.988 $\approx$ 2**	dimensionless
EM term	a_EM	4.136×10^{-10}	m/s^2

UQFF computed: Canonical UQFF buoyancy parameter U_bi = ?�[SSq]�GM/rκ = 5.0e-4�0.57�6.67e-11�M/r�; for solar parameters: U_bi,Sun = 5.7e-4�6.67e-11�1.99e30/(6.96e8)� = 1.47e+2 m/s�.

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_\mu \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac,[SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac,[SCm]}} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2\phi_{\text{NS}} - (4\pi G\rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}}\partial_t\phi_{\text{NS}} = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{\text{4 forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S/\delta\phi_{\text{NS}} =$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac,[SCm]}}/\rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac,[SCm]}} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.116 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36}$ kg/m^3.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 107, \quad n_{\text{channel}} = 12/26$$

Since $p_{\text{DVP}} = 107$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f'_{\text{UA}} + f_{\text{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is $\mathbf{10^4}$ **yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq]\cdot m/M_\odot}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.116	✓ Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\text{DVP}} = 107$	✓ Resonant
BSH layers	26 harmonic terms	j = 1...26, $\cos(2\pi j/26)$	✓ Full 26D projection

Framework	Canonical Value	This Paper	Status
κ decay	5.0×10^{-4} day^{-1}	Applied in VDS exponential	✓ Canonical
[SSq]	0.57	Applied in BSH saturation	✓ Canonical